

SALES & INVENTORY ANALYTICS FOR SPINNER PLASTIC INDUSTRIES LIMITED



PROJECT REPORT

Submitted to

**School of Management and Business Studies,
Mahatma Gandhi University**

Submitted by

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in partial fulfilment of the requirements for the award of the Degree of

MASTER OF BUSINESS ADMINISTRATION



**School of Management and Business Studies
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July 2026

DECLARATION

I, **Riyas Sakeer** student of the School of Management and Business Studies, hereby declare that the project titled "**Sales & Inventory Analytics for Spinner Plastic Limited**" submitted in partial fulfilment of the requirements for the award of the Master of Business Administration from Mahatma Gandhi University is a bona fide record of work done by me. This report has not previously formed the basis for the award of any degree, diploma, or similar title of any University.

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CERTIFICATE

This is to certify that **Mr. RIYAS SAKEER** Student of School of Management and Business Studies Mahatma Gandhi university, Kottayam (Registration No. **MG24B1090054**) has completed project on the topic "**Sales & Inventory Analytics for Spinner Plastic Industries Limited**" at Spinner Plastic Industries Limited, Athani, Thrissur, from 16-04-2026 to 12-06-2026.

He was very inquisitive during his project period and was eager to know overall functions of our organization. During the course of his work, he was found very promising and acquired excellent knowledge on the topic.

For Spinner Plastic Industries Limited,


P.J. GEORGEKUTTY

(Managing Director)



13-06-2026

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CERTIFICATE

*This is to certify that the report titled “**Sales and Inventory Analytics for Spinner Plastic Industries Limited**” is a bona fide record of the project work done by **Riyas Sakeer** under the guidance of **Dr. Madhu Lal M**, in partial fulfilment of the requirements for the award of the Degree of Master of Business Administration of Mahatma Gandhi University, Kottayam, Kerala.*

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ACKNOWLEDGEMENT

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Riyas Sakeer

EXECUTIVE SUMMARY

The present study titled "Sales and Inventory Analytics for Spinner Plastic Industries Limited" examines the sales performance, inventory efficiency, cost structure, gross margin, regional distribution, and financial health of Spinner Plastic Industries Limited, Thrissur, Kerala, for the period 2022 to 2025 for sales data and FY 2020–21 to FY 2022–23 for financial data.

The study was undertaken to analyse product-wise sales trends, evaluate inventory turnover using ABC classification, compare gross margin between manufacturing and trading segments, examine cost of production structure, assess regional sales distribution across Kerala, and analyse financial performance through Profit & Loss and Balance Sheet data. Primary data were collected through discussions with company management and secondary data from financial statements, catalogues, industry reports, and online sources. Tools such as trend analysis, ABC classification, gross margin analysis, seasonal index analysis, and hypothesis testing were employed using Microsoft Excel and Power BI Desktop.

The findings revealed consistent revenue growth from ₹321 million in 2022 to ₹370 million in 2025 at 5 per cent annually. uPVC Pipes (26%) and HDPE Pipes (21%) were the top contributors. ABC classification identified these along with Rain Water Gutters as Category A items with 15–18 day stock turnover, while trading products recorded holding periods of 85–90 days. Seasonal analysis confirmed peak months (October–May) averaging a sales index of 111.25 against 67.50 in off-peak months. Regional analysis showed zero sales in Trivandrum and Kollam, with over 92 per cent of sales concentrated in North and Central Kerala. Gross profit margin remained stable at 18.57 per cent and net margin at 5.00 per cent across FY 2020–2023.

The study concludes that Spinner Plastic Industries Limited is financially stable with consistent growth and sound debt management. Adopting ABC-based inventory reviews, expanding dealer presence in South Kerala, and monitoring debtor levels are essential for enhancing operational efficiency and long-term growth.

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

In today's knowledge-driven economy, data analytics has emerged as a transformative force across all industries, enabling organisations to move beyond intuition-based management towards evidence-driven decision making. Manufacturing firms, in particular, stand to gain significantly from adopting analytics practices, as they deal with large volumes of operational, financial, and market data on a daily basis. The ability to interpret this data effectively determines a company's capacity to optimise its resources, respond to market signals, and sustain competitive advantage. The Indian plastic manufacturing industry is one of the most dynamic and rapidly growing sectors in the country. Valued at approximately USD 36.14 billion in 2023, the market is projected to reach USD 56.62 billion by 2035, growing at a compound annual growth rate (CAGR) of 3.81% (Mordor Intelligence, 2024). This growth is driven by increasing demand from sectors such as agriculture, construction, telecommunications, and consumer goods. Within Kerala, approximately 1,340 plastic processing units operate, predominantly Micro, Small, and Medium Enterprises (MSMEs), collectively employing an estimated 100,000 people (Kerala State Planning Board, 2024).

Despite this growth, many small and medium-scale manufacturers in Kerala have not yet adopted structured analytics frameworks for managing their sales, inventory, and financial performance. This study seeks to address that gap by applying business analytics tools to real operational data from Spinner Plastic Industries Limited, a Thrissur-based manufacturer with over three decades of operational history.

1.2 INDUSTRY AND COMPANY PROFILE

1.2.1 Indian Plastic Pipe Industry

The plastic pipe industry in India occupies a central position within the broader plastics manufacturing ecosystem. Plastic pipes are extensively used in

agricultural irrigation, water supply and distribution systems, sewerage and drainage, telecommunication cable ducting, and industrial fluid handling. HDPE (High-Density Polyethylene) and UPVC (Unplasticised Polyvinyl Chloride) pipes dominate the market due to their durability, corrosion resistance, lightweight properties, and cost-effectiveness compared to traditional materials such as iron and concrete.

India's plastic pipe market has witnessed consistent growth over the past decade, supported by government infrastructure initiatives such as the Jal Jeevan Mission (targeting household tap water connections), Smart Cities Mission, PMGSY rural road connectivity schemes, and large-scale irrigation projects. The demand is expected to grow further with increasing urbanisation and agricultural modernisation across India (FICCI, 2023).

Kerala's plastic manufacturing industry is concentrated in districts such as Ernakulam, Thrissur, and Kozhikode. Southern India accounts for approximately 21% of India's total plastic consumption. Key challenges for the industry include volatile raw material prices (primarily PVC resin and HDPE granules, which are petrochemical derivatives), environmental regulations, and competition from large national brands. Nevertheless, regional manufacturers with strong distribution networks and brand recognition continue to hold significant market share in their local geographies.

1.2.2 Company Profile: Spinner Plastic Industries Limited

Spinner Plastic Industries Limited is a well-established manufacturer of plastic pipe systems and sanitary ware, headquartered at Athani Post Office, Thrissur, Kerala — 680581. The company was originally established in 1992 as Peejay Industries, focusing initially on steel power transmission equipment. In 1996, it strategically diversified into plastic pipe manufacturing, commencing production of LDPE pipes, and subsequently expanded into HDPE, UPVC, and

Soft PVC pipe segments. This diversification led to the establishment of Spinner Polymers Pvt. Ltd. and Spinner Extrusions Pvt. Ltd. as group entities.

The company operates under the flagship brand 'SPINNER' and holds ISI certification for its pipe products, adhering to BIS (Bureau of Indian Standards) quality norms. Spinner maintains an in-house Quality Control Laboratory and Testing Facility, ensuring product compliance with both ISI and international standards. The company has received recognition from various governmental and industrial bodies, including credit rating certifications and entrepreneurship awards.

The current product portfolio of Spinner Plastic Industries Limited includes:

- HDPE ISI Pipes and Non-ISI Pipes (used in water supply, irrigation, and telecommunications)
- LDPE Pipes (used in low-pressure agricultural and domestic applications)
- UPVC Pipes (used in electrical conduit, plumbing, and civil applications)
- Soft PVC Pipes (flexible pipes for garden, industrial, and domestic use)
- Rainwater Gutters (used in residential and commercial construction)
- Water Storage Tanks (LLDPE-based tanks in capacities from 200L to 1000L)
- Drip Irrigation Pipes and Fittings
- Ceramic and Sanitary Ware (closets, wash basins, urinals — trading products)
- Power Transmission Equipment (trading products supplied to KSEB and similar clients)

The company's vision is to be a market leader and innovative provider of high-quality piping solutions, prioritising customer satisfaction, continuous product improvement, and adherence to safety and environmental standards. The manufacturing facility at Thrissur serves as both the production hub and the primary distribution point for its dealer network spread across Kerala.

1.3 NEED AND SIGNIFICANCE OF THE STUDY

Manufacturing SMEs in Kerala often manage operations through experience and intuition rather than systematic data analysis. While this approach may have sufficed in earlier decades, the increasingly competitive market environment — characterised by volatile raw material costs, shifting seasonal demand patterns, and expanding product portfolios — demands a more structured, data-driven approach to business management.

Spinner Plastic Industries Limited, despite its strong market presence in North and Central Kerala, operates in a segment that requires careful management of inventory, sales strategy, and financial resources. A structured analytics study can provide the management with a clear picture of which products are driving revenue, which inventory items are causing working capital blockage, which regions offer untapped growth potential, and how the company's financial position has evolved over recent years.

From an academic perspective, this study contributes empirical evidence to the field of business analytics applied to manufacturing SMEs in Kerala, a domain that remains under-researched. The findings enrich the understanding of analytics adoption in non-metropolitan industrial contexts and provide a replicable framework for similar companies. From a managerial perspective, the study equips Spinner's management with actionable insights, dashboards, and visual analytics outputs that can directly support strategic decisions on product focus, dealer network expansion, inventory rationalisation, and financial planning.

1.4 STATEMENT OF THE PROBLEM

Spinner Plastic Industries Limited manages a diverse portfolio of nine product categories across manufacturing and trading segments, serves dealers across multiple districts of Kerala, and operates across peak and off-peak demand cycles. However, in the absence of a formal analytics framework, the management lacks structured visibility into key performance indicators such as product-wise sales contribution, inventory turnover rates, gross margin differentials between manufacturing and trading products, regional sales concentration, and financial performance trends.

This study attempts to address the following core questions:

- Which product categories contribute most to sales revenue and which are underperforming?
- How are inventory levels and stock holding periods distributed across product lines, and which products are slow-moving?
- What is the gross margin differential between manufacturing and trading products, and how has it trended over time?
- How is the company's sales geographically distributed across Kerala, and which districts represent growth opportunities?
- What does the financial performance data (P&L and Balance Sheet) reveal about the company's profitability and financial health over the study period?

1.5 OBJECTIVES OF THE STUDY

The primary objectives of this study are:

- To analyse product-wise and category-wise sales trends of Spinner Plastic Industries Limited over the period 2022–2025.
- To evaluate inventory turnover and identify fast-moving and slow-moving products using ABC inventory classification.

- To compare gross margin performance between manufacturing and trading product segments.
- To examine the cost of production structure across product lines, differentiating between raw material, manufacturing, and trading expense components.
- To assess regional sales distribution across Kerala districts and identify geographic concentration and gaps.
- To analyse the financial performance of the company through Profit & Loss and Balance Sheet data for the period 2020–2023.
- To provide data-driven recommendations to the management of Spinner Plastic Industries Limited based on the findings of the analysis.

1.6 SCOPE OF THE STUDY

The study is confined to Spinner Plastic Industries Limited, Thrissur, Kerala. The analysis covers the following dimensions:

- Product-wise sales data for the period 2022 to 2025 across nine product categories.
- Inventory data and ABC classification for the period 2022 to 2024.
- Cost of production data covering raw material, manufacturing, and trading expense components.
- Gross margin analysis across product categories and years.
- Regional (district-wise) sales distribution across 14 districts of Kerala.
- Seasonal demand patterns based on monthly sales index data.
- Financial performance data from the company's Profit & Loss Account and Balance Sheet for the period FY 2020 to FY 2023.

The study does not extend to competitor benchmarking, customer satisfaction measurement, or employee performance assessment. The analysis is based on data provided by the company, including actual figures, approximate values, and company-stated percentages, which have been structured and organised by the researcher for the purpose of this study.

1.7 LIMITATIONS OF THE STUDY

The following limitations are acknowledged:

- The study is confined to a single organisation — Spinner Plastic Industries Limited, Thrissur, Kerala — and the findings cannot be generalised to other plastic manufacturing companies or to the broader plastic pipe industry in India or Kerala.
- The financial data analysed covers only the period FY 2020 to FY 2023, and more recent financial statements were not available for inclusion, which limits the ability to comment on the company's most current financial position.
- The study does not include primary data collected through surveys, interviews, or questionnaires from customers, dealers, or employees. The absence of stakeholder perspectives limits the depth of qualitative insight that could otherwise complement the quantitative findings.
- The research does not analyse competitor performance or benchmark Spinner's metrics against industry peers. Without such comparative data, it is not possible to determine whether the company's growth rate, margins, or inventory levels are above or below industry standards.
- The study does not examine customer satisfaction, dealer feedback, or after-sales service quality — dimensions that directly influence long-term sales performance but fall outside the scope of this analytics-focused project.

- The analysis does not cover human resource metrics, energy efficiency indicators, or environmental compliance data, which are increasingly relevant to manufacturing SMEs operating under evolving regulatory frameworks in Kerala.
- The study covers sales data for the period 2022 to 2025 and financial statements for FY 2020 to FY 2023, resulting in a difference in the time periods covered across the two data dimensions, which limits direct year-on-year comparison between sales performance and financial outcomes.

1.8 ORGANISATION OF THE REPORT

Chapter 1 — Introduction: Presents the background, company and industry profile, need and significance of the study, problem statement, objectives, scope, and limitations.

Chapter 2 — Theoretical Framework: Reviews relevant academic literature and theoretical concepts underpinning the analytical tools and frameworks used in the study.

Chapter 3 — Research Methodology: Describes the research design, data sources, tools and techniques used for analysis, and the hypothesis tested.

Chapter 4 — Data Analysis and Interpretations: Presents the analytical findings across all study dimensions with supporting visual charts and interpretations.

Chapter 5 — Findings, Suggestions, and Conclusion: Summarises the key findings, offers managerial recommendations, and concludes the study.

CHAPTER 2

THEORETICAL FRAMEWORK

2.1 SALES ANALYTICS

Sales analytics refers to the practice of using quantitative methods to understand, predict, and improve the sales performance of an organisation. It encompasses the collection, processing, and interpretation of sales data to identify patterns, trends, and opportunities (Witten, Frank, Hall, & Pal, 2016). In manufacturing firms, sales analytics helps management understand which products are contributing to revenue growth, which markets are underperforming, and how seasonal cycles affect demand.

Davenport and Harris (2007) define analytics as the extensive use of data, statistical and quantitative analysis, explanatory and predictive models, and fact-based management to drive decisions and actions. For SMEs in developing markets, even basic descriptive analytics — trend analysis, percentage distribution analysis, and comparative ranking — can yield significant strategic value (Kiron et al., 2014).

2.2 INVENTORY MANAGEMENT AND ABC CLASSIFICATION

Inventory management is a critical operational function in manufacturing organisations, directly influencing working capital efficiency, production continuity, and customer service levels. Excess inventory ties up capital and increases storage costs, while insufficient inventory leads to stockouts and lost sales (Chase, Aquilano, & Jacobs, 2006).

The ABC classification system is one of the most widely used inventory management tools. Originally derived from the Pareto Principle (also known as the 80-20 rule), it classifies inventory items into three categories based on their annual consumption value (Tersine, 1994):

- Category A: High-value items that account for approximately 60–70% of total inventory value but represent only 10–20% of total items. These require tight control, frequent review, and accurate forecasting.

- Category B: Medium-value items accounting for 20–30% of inventory value, requiring moderate control with periodic review.
- Category C: Low-value items that account for only 5–10% of inventory value but represent a large proportion of total items. These can be managed with simpler, less frequent review mechanisms.

In the context of Spinner Plastic Industries Limited, ABC classification enables management to identify which pipe product categories demand priority attention in terms of procurement planning, reorder point setting, and stock level maintenance.

Inventory turnover ratio measures how many times a company's inventory is sold and replenished within a year, calculated by dividing 365 by the average inventory holding period in days (Chase, Aquilano, & Jacobs, 2006).

A high turnover ratio indicates efficient stock movement and healthy demand, while a low ratio signals slow-moving inventory and working capital blockage. Products with turnover ratios below 12 times per year — equivalent to a 30-day holding period — are considered inefficient and require managerial intervention through demand stimulation or order reduction.

2.3 COST OF PRODUCTION ANALYSIS

Cost of production refers to the total expenditure incurred in producing goods, encompassing raw material costs, manufacturing overhead, and other production-related expenses (Horngren, Datar, & Rajan, 2012). In the plastic pipe manufacturing industry, raw materials such as PVC resin and HDPE granules constitute the largest cost component, typically accounting for 65 to 70 per cent of selling price. For companies operating both manufacturing and trading segments, the cost structure differs significantly — trading products do not incur production costs and instead carry purchase cost and trading expenses

as their primary cost components (Narayanan & Sarkar, 2002). A structured cost of production analysis across product lines enables management to identify cost concentration, compare segment-level profitability, and make informed decisions on pricing and resource allocation.

2.4 GROSS MARGIN ANALYSIS

Gross margin is defined as the difference between net sales revenue and the cost of goods sold (COGS), expressed as a percentage of net sales. It is a primary indicator of a company's pricing efficiency and production cost control (Horngren, Datar, & Rajan, 2012). For a company like Spinner, which operates both a manufacturing segment (where COGS includes raw materials and manufacturing overheads) and a trading segment (where COGS primarily consists of purchase cost of goods), gross margin analysis by product segment provides critical insight into the profitability structure of the business.

Research by Narayanan and Sarkar (2002) demonstrates that companies with segmented cost and margin visibility are better positioned to make strategic decisions about product portfolio rationalisation, pricing adjustments, and channel profitability. Understanding the gross margin differential between manufacturing and trading products allows Spinner to assess the relative profitability contribution of each segment and prioritise resource allocation accordingly.

2.5 SEASONAL DEMAND ANALYSIS

Seasonal demand patterns are a common feature of industries connected to agricultural cycles, construction activity, or weather-driven consumption. In the context of plastic pipe manufacturing in Kerala, demand is significantly influenced by agricultural and construction activity, which tends to peak in the post-monsoon and pre-monsoon seasons (October to May) and slow during the monsoon period (June to September).

Chopra and Meindl (2016) note that supply chain planning in industries with pronounced seasonality must account for demand variability through strategic inventory positioning, production smoothing, and flexible procurement arrangements. A seasonal demand index — which measures monthly demand as a proportion of the annual average — is a practical tool for visualising and communicating these patterns to management, enabling more informed production scheduling and dealer stocking decisions.

2.6 REGIONAL SALES ANALYSIS

Geographic sales analysis is a form of market intelligence that maps sales performance across territories to identify areas of concentration, under-penetration, and potential growth. Kotler and Keller (2016) describe geographic segmentation as an essential component of market analysis, particularly for companies with regional distribution networks.

For Spinner Plastic Industries Limited, regional sales analysis across Kerala districts provides management with a clear picture of geographic market concentration (North and Central Kerala) and identifies southern districts as potential expansion opportunities. District-level sales index mapping, as used in this study, is a practical approach adapted from regional sales performance tracking used in FMCG and industrial distribution management.

2.7 FINANCIAL PERFORMANCE ANALYSIS

Financial statement analysis is a fundamental tool in business management and academic research for assessing organisational health, efficiency, and sustainability. The primary financial statements used in this study — the Profit & Loss Account and the Balance Sheet — provide a structured view of revenue generation, cost structure, profitability, asset composition, and financial leverage (Pandey, 2009).

Key financial metrics examined in this study include gross profit margin, net profit margin, fixed asset trends (with depreciation), loan reduction trajectory, and reserve and surplus growth. Brigham and Ehrhardt (2014) note that trend analysis of financial statements over multiple years reveals more meaningful insights than single-period snapshots, as it captures the directional movement of key performance indicators.

2.8 BUSINESS INTELLIGENCE AND DATA VISUALISATION

Business intelligence (BI) refers to technologies, applications, and practices for the collection, integration, analysis, and presentation of business information (Negash, 2004). Power BI, developed by Microsoft, is a leading BI tool that enables analysts to connect to multiple data sources, build interactive dashboards, and generate publication-quality visual reports. For SMEs, BI tools provide a cost-effective means of converting raw operational data into actionable visual insights accessible to non-technical managers.

Few and Grant (2014) argue that effective data visualisation reduces cognitive load and enables faster. In this study, Power BI has been used as the primary tool for building all analytical charts and dashboards, with outputs formatted for inclusion in this academic report.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 OBJECTIVES OF THE STUDY

The study aims to analyse the sales performance, inventory efficiency, cost structure, gross margin, regional distribution, and financial health of Spinner Plastic Industries Limited using business analytics tools, with the objective of providing data-driven insights and recommendations to the management.

3.2 HYPOTHESIS

Based on the nature of the study, the following hypothesis has been formulated:

H₀ (Null Hypothesis): There is no significant difference in the sales performance of Spinner Plastic Industries Limited across peak and off-peak seasons.

H₁ (Alternate Hypothesis): There is a significant difference in the sales performance of Spinner Plastic Industries Limited across peak and off-peak seasons.

This hypothesis is tested using the seasonal demand index data, comparing the average sales index during the eight peak months (October–May) against the four off-peak months (June–September).

3.3 RESEARCH DESIGN

This study follows a descriptive and analytical research design. Descriptive research is used to systematically describe the characteristics, trends, and patterns in the company's sales, inventory, cost, and financial data. Analytical research extends this by examining relationships between variables, comparing performance across dimensions (product, region, season, year), and drawing interpretive conclusions.

The study is quantitative in nature, relying on numerical data structured into datasets for analysis using Power BI as the primary visualisation and analytics tool.

3.4 SOURCES OF DATA

3.4.1 Primary Data

Primary data was collected directly from Spinner Plastic Industries Limited through interactions with the management and company representatives. The data obtained includes:

- Approximate total annual sales figures (FY 2020 actual)
- Product-wise sales distribution percentages across nine product categories
- Inventory data: approximate monthly production volumes, reorder levels, and stock holding periods for fast-moving and slow-moving products
- Cost of production percentages: raw material cost (65–70%), manufacturing overhead (16–20%) for manufacturing products; purchase cost (70%) and trading expenses (5–10%) for trading products
- Regional sales information: top-selling districts, no-sale districts, and approximate sales distribution
- Seasonal demand patterns: peak season (October–May) and off-peak season (June–September)
- Annual sales growth rate: 5% year-on-year increase

3.4.2 Secondary Data

Secondary data was obtained from the following sources:

- Published annual financial statements of Spinner Plastic Industries Limited (FY 2020 actual figures from company documents)
- Company product catalogue and price lists (Spinner Pipes official catalogue)
- Industry reports on the Indian plastic pipe market (Mordor Intelligence, FICCI, Kerala State Planning Board)
- Academic literature on sales analytics, inventory management, and financial analysis

3.5 SAMPLING METHOD

This study does not involve survey-based sampling of respondents. The unit of analysis is the company — Spinner Plastic Industries Limited — and the data analysed represents the entire operational scope of the company across the study period. All available data dimensions (products, districts, years, financial accounts) are included in the analysis, making this a census-based secondary data study rather than a sample-based primary data study.

3.6 METHOD OF DATA COLLECTION

Data was collected through:

- Direct discussion with company management and the company guide during the project tenure
- Review of company documents including the company catalogue, price list, and financial records
- Structured data elicitation: the researcher asked specific, targeted questions to obtain percentage distributions, approximate volumes, and cost ratios, which were then structured into an organised Excel dataset.

3.7 TOOLS USED FOR ANALYSIS

The following tools were used for data analysis and visualisation:

- Microsoft Excel (for dataset construction, structuring of 10 data tables, and preliminary calculations)
- Microsoft Power BI Desktop (for building all analytical charts, dashboards, and visual reports)

The dataset was organised into ten structured tables:

Table 3.1: Datasets List

Sheet Name	Contents
Sales Data	Monthly product-wise sales (₹), 2022–2025
Sales Data Unpivoted	Unpivoted format for product-level visualisation
Cost of Production	Year-wise cost breakdown by product and type
Gross Margin Analysis	Product-wise gross margin (₹ and %) by year
Inventory Data	Stock levels, reorder levels, holding periods
Region Sales	District-wise sales index and share (%)
Profit & Loss	P&L account data, FY 2020–2023
Balance Sheet	Balance sheet data, FY 2020–2023
Seasonal Trend	Monthly sales index — peak and off-peak
ABC Classification	ABC category, inventory days, value share

Source: Compiled by the researcher

3.8 DATA ANALYSIS TECHNIQUES

The following analytical techniques are applied in this study:

- Trend Analysis: Examining year-over-year changes in sales, costs, margins, and financial indicators.
- Percentage Distribution Analysis: Calculating the proportional share of each product, region, or cost component in the total.
- ABC Inventory Classification: Ranking products by annual sales value contribution and classifying into A, B, and C categories.
- Gross Margin Analysis: Computing gross margin (₹ and %) for each product, differentiating manufacturing from trading cost structures.

- Seasonal Index Analysis: Building a monthly sales index (base = 100) to quantify peak and off-peak demand variation.
- Comparative Analysis: Comparing performance across product categories, regions, years, and financial periods.
- Financial Statement Analysis: Analysing Profit & Loss and Balance Sheet data to assess profitability, cost structure, loan reduction, and asset composition.

All visual outputs are generated in Microsoft Power BI and are presented as figures in Chapter 4.

CHAPTER 4

**DATA ANALYSIS AND
INTERPRETATIONS**

4.1 SALES TREND ANALYSIS

This section analyses the sales performance of Spinner Plastic Industries Limited across the period 2022–2025. The analysis covers year-wise total sales, monthly demand distribution, and product-wise revenue contribution.

4.1.1 Year-wise Total Sales (2022–2025)

Figure 4.1 presents the year-wise total sales trend of Spinner Plastic Industries Limited from 2022 to 2025. The company recorded total sales of approximately ₹321 million in 2022, which increased to ₹338 million in 2023 and ₹356 million in 2024, before reaching approximately ₹370 million in 2025. This reflects a consistent year-on-year growth of approximately 5%, in line with the company's stated growth trajectory. The sustained upward trend indicates healthy revenue growth and confirms the company's ability to expand its market reach over the study period.

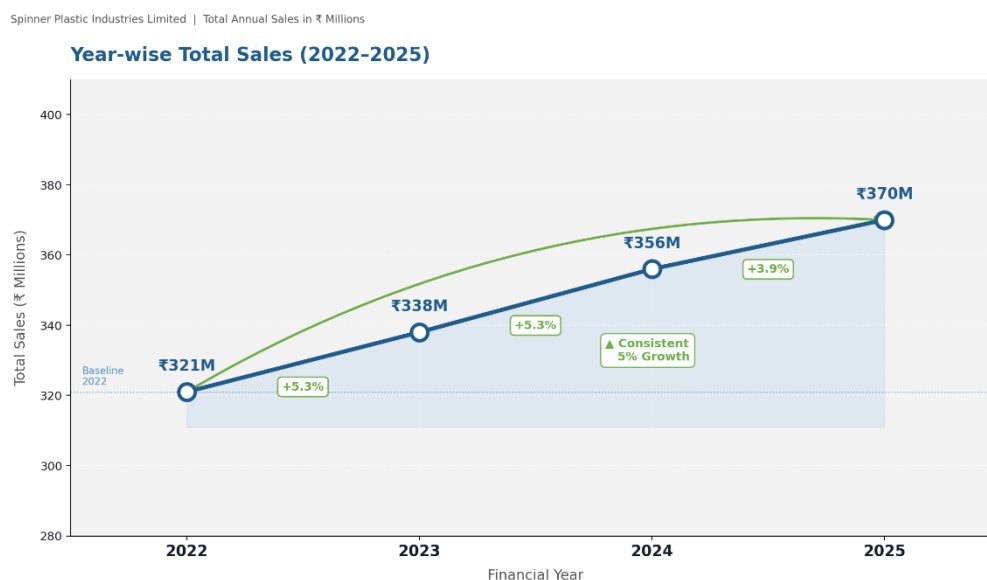


Figure 4.1: Year-wise Total Sales (2022–2025)

Source: Sales Records

4.1.2 Monthly Sales Trend by Year

Figure 4.2 presents the month-wise sales distribution for each year from 2022 to 2025. The chart clearly demonstrates the seasonal demand pattern of the company. Sales are consistently higher during the October–May period, with peak months recording values between ₹30 million and ₹37 million per month. The June–September period reflects a significant dip, with monthly sales falling to the ₹18–23 million range. This pattern is consistent across all four years, confirming the pronounced seasonality of demand. March and October–November consistently emerge as the highest-sales months, driven by pre-monsoon construction activity and post-monsoon agricultural demand respectively.

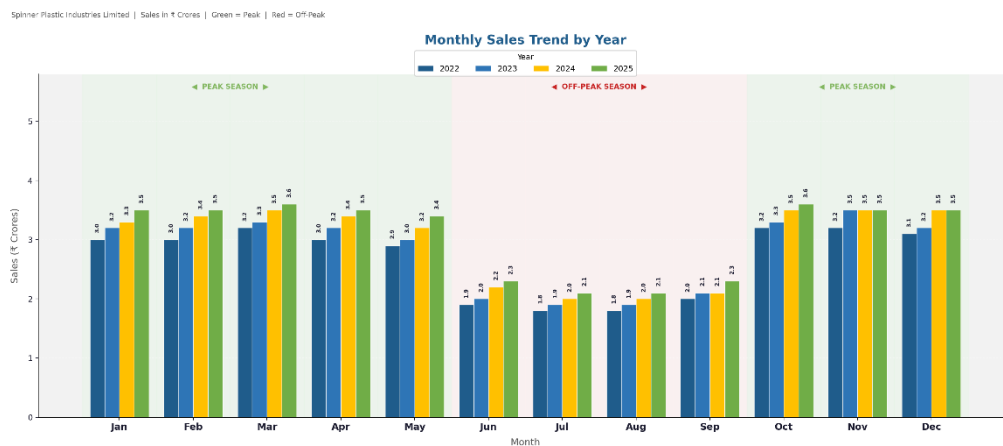


Figure 4.2: Monthly Sales Trend by Year (2022–2025)

Source: Monthly Sales Records

4.1.3 Product-wise Sales by Year (Stacked Bar)

Figure 4.3 shows the product-wise composition of annual sales from 2022 to 2025 in a stacked bar format. uPVC Pipes and HDPE Pipes collectively dominate sales, accounting for approximately 45–47% of total revenue in each year. Rain Water Gutters represent the third-largest contributor. The composition has remained broadly consistent across years, indicating stable product demand ratios. Minor shifts in the contribution of individual products are observed, reflecting marginal changes in market mix and product availability.

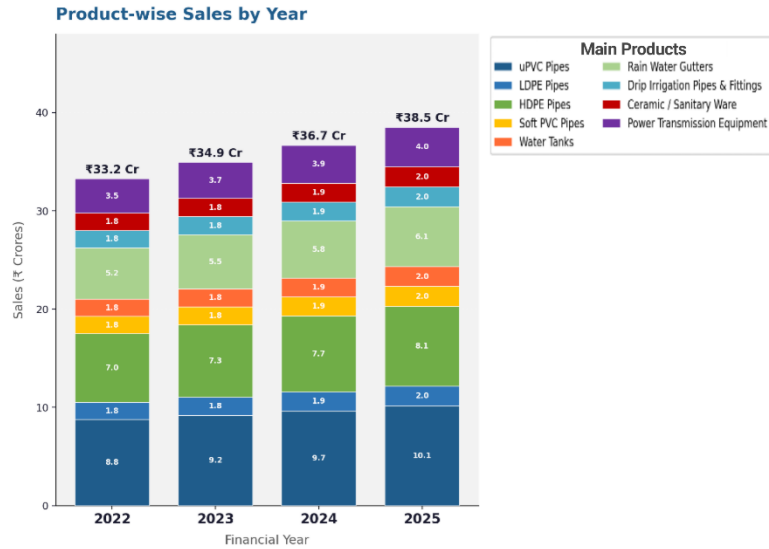


Figure 4.3: Product-wise Sales by Year — Stacked Bar (2022–2025)

Source: Product -wise Sales Records

4.1.4 Product Sales Distribution (2025)

Figure 4.4 presents the product sales distribution for 2025 as a pie chart. uPVC Pipes lead with a 26% share, followed by HDPE Pipes at 21%, Rain Water Gutters at 16%, and Power Transmission Equipment at 10%. The remaining five product categories — LDPE Pipes, Soft PVC Pipes, Water Tanks, Drip Irrigation Pipes & Fittings, and Ceramic/Sanitary Ware — each account for approximately 5% of total sales. This distribution underscores the company's dependence on pipe products as the primary revenue drivers, while the trading segments (Sanitary Ware and Power Transmission Equipment) represent relatively smaller but not insignificant contributions.

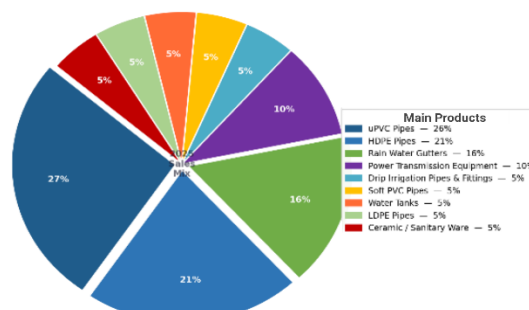


Figure 4.4: Product Sales Distribution — 2025 (Pie Chart)

Source: Product -wise Sales Records

4.1.5 Category-wise Sales Trend (2022–2025)

Figure 4.5 presents the category-wise sales trend of Spinner Plastic Industries Limited for the period 2022 to 2025, classifying the company's product portfolio into two broad categories — Manufacturing and Trading. The Manufacturing category, which comprises HDPE Pipes, uPVC Pipes, LDPE Pipes, Soft PVC Pipes, Rainwater Gutters, Water Tanks, and Drip Irrigation Pipes and Fittings, accounted for approximately 95 per cent of total annual sales throughout the study period, recording sales of ₹305 million in 2022 and growing consistently to ₹352 million in 2025, reflecting the 5 per cent annual growth trajectory of the company. The Trading category, which includes Ceramic and Sanitary Ware and Power Transmission Equipment, contributed approximately 5 per cent of total sales, recording ₹16 million in 2022 and reaching ₹18 million in 2025.

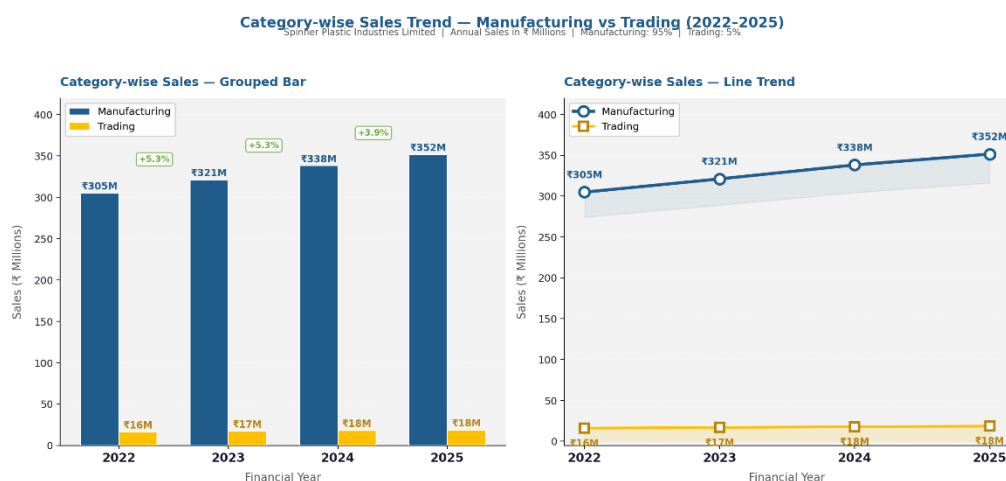


Figure 4.5: Category-wise Sales Trend -Manufacturing vs. Trading(2022-2025)

Source: Product -wise Sales Records

4.2 PRODUCT PERFORMANCE ANALYSIS

This section evaluates individual product performance in terms of sales ranking, growth trajectory, and revenue share.

4.2.1 Product-wise Sales Ranking

Figure 4.6 ranks all nine product categories by average sales share. uPVC Pipes lead with 26.10%, followed by HDPE Pipes (21.39%), Rain Water Gutters (15.68%), Power Transmission Equipment (10.46%), and Drip Irrigation Pipes & Fittings (5.32%). The bottom four products — Soft PVC Pipes, Water Tanks, LDPE Pipes, and Ceramic/Sanitary Ware — each account for approximately 5% of total sales. This ranking provides clear prioritisation guidance for the sales and production teams in terms of resource allocation, stock management, and market focus.

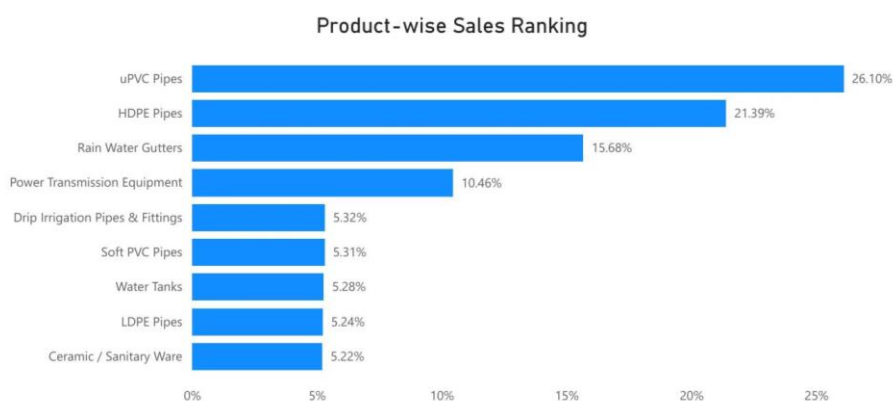


Figure 4.6: Product-wise Sales Ranking

Source: Product -wise Sales Records

4.2.2 Product-wise Sales Growth Trend (2022–2025)

Figure 4.7 tracks the sales growth of key products from 2022 to 2025. uPVC Pipes show the strongest absolute growth, increasing from ₹84 million in 2022 to approximately ₹96 million in 2025, before showing a slight decline in 2024 and 2025. HDPE Pipes and Rain Water Gutters show a broadly positive growth trajectory. Water Tanks and Soft PVC Pipes remain relatively flat, consistent with their low-growth, slow-moving classification. The growth trend broadly follows the overall 5% annual growth rate, though individual product variations reflect market-specific dynamics.

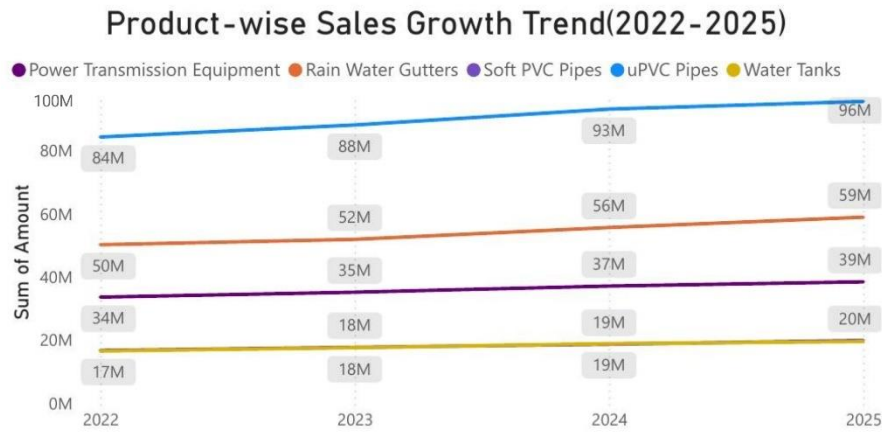


Figure 4.7: Product-wise Sales Growth Trend (2022–2025)

Source: Product -wise Sales Records

4.2.3 Product Revenue Share (Donut Chart)

Figure 4.8 presents the product revenue share in donut chart format, providing a clear visual representation of the proportional contribution of each product category to total sales. The chart reinforces the finding that the top three products — uPVC Pipes, HDPE Pipes, and Rain Water Gutters — collectively account for approximately 63% of total revenue, while the remaining six categories share the remaining 37%.

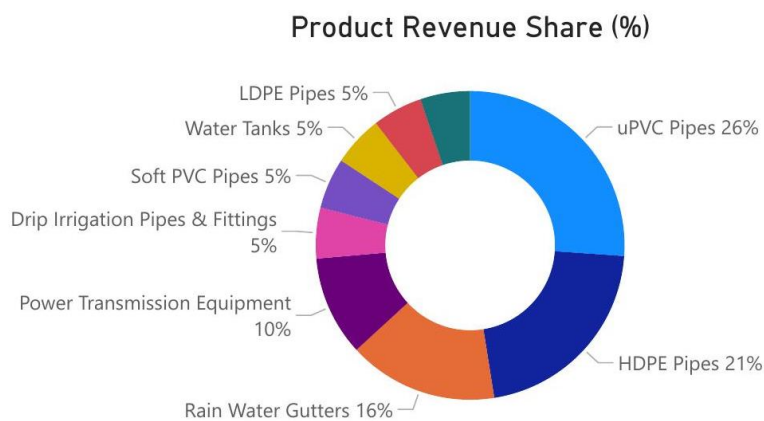


Figure 4.8: Product Revenue Share (%)

Source: Product -wise Sales Records

4.3 COST OF PRODUCTION ANALYSIS

Figure 4.9 presents the cost of production structure across all nine product lines of Spinner Plastic Industries Limited, differentiating between raw material cost, manufacturing overhead, trading expenses, and gross margin components as a percentage of sales revenue. For all seven manufacturing products — uPVC Pipes, HDPE Pipes, LDPE Pipes, Soft PVC Pipes, Rainwater Gutters, Water Tanks, and Drip Irrigation Pipes and Fittings — the cost structure is uniform, with raw material accounting for 67.5 per cent of sales revenue and manufacturing overhead contributing 18.0 per cent, resulting in a total production cost of 85.5 per cent and a gross margin of 14.5 per cent.. For the two trading products — Power Transmission Equipment and Ceramic and Sanitary Ware — the cost structure is fundamentally different, comprising a purchase cost of 70.0 per cent and trading expenses of 7.5 per cent, totalling 77.5 per cent, and leaving a higher gross margin of 22.5 per cent. The cost structure analysis clearly illustrates that while manufacturing products bear a heavier cost burden due to raw material and production overhead, trading products achieve higher margins by eliminating production-related costs entirely and operating purely as a buy-and-sell model.

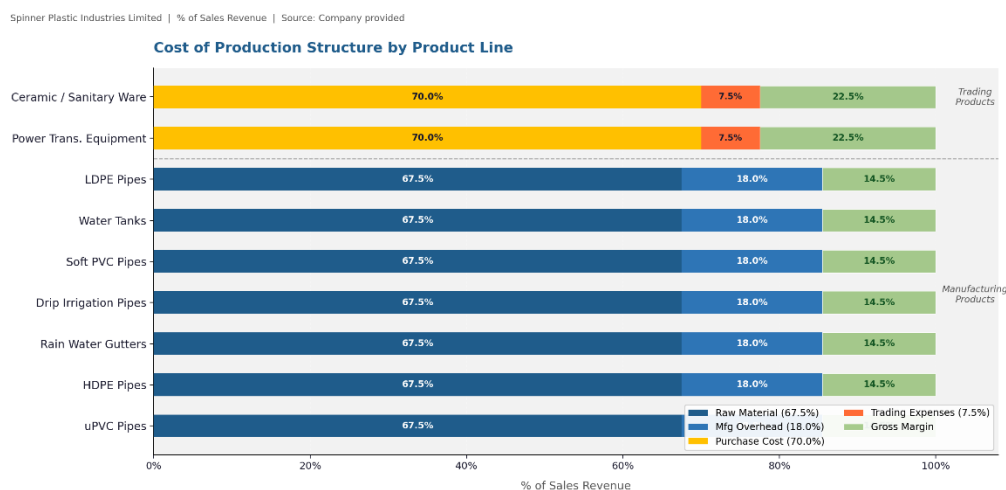


Figure 4.9: Cost of Production Structure by Product Line

Source: Production Cost Records

4.4 MANUFACTURING VS. TRADING ANALYSIS

This section compares the revenue contribution and gross margin performance of manufacturing products (which the company produces in-house) against trading products (which are procured and resold).

4.4.1 Manufacturing vs. Trading Revenue Split

Figure 4.10 presents the revenue split between manufacturing and trading segments. Manufacturing products account for approximately 95% of total revenue, while trading products (Ceramic/Sanitary Ware and Power Transmission Equipment combined) contribute approximately 5%. This confirms that Spinner Plastic Industries Limited is fundamentally a manufacturing company, with trading representing a supplementary revenue stream rather than a core business.

Manufacturing vs Trading Revenue Split

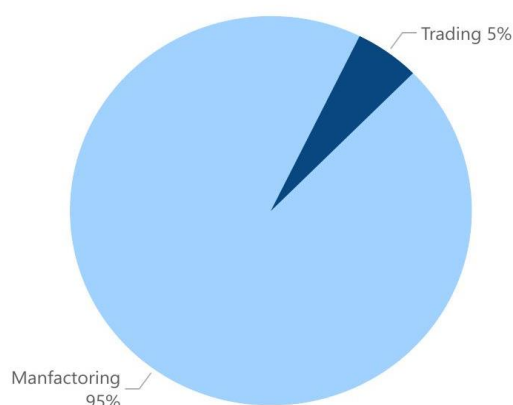


Figure 4.10: Manufacturing vs. Trading Revenue Split

Source: Sales Records

4.4.2 Gross Margin % — Manufacturing vs. Trading

Figure 4.11 compares the gross margin percentage between manufacturing and trading segments. Trading products record a gross margin of 22.50%, while manufacturing products record 14.50%. The higher gross margin for trading

products is attributable to the fact that trading involves purchasing finished goods (at approximately 70% of selling price plus 7.5% trading expenses, totalling 77.5% cost) and reselling at a margin, without incurring the heavier cost burden of raw material processing and manufacturing overhead that applies to manufactured products (85.5% total cost: 67.5% raw material + 18% manufacturing). Despite the higher margin percentage, the absolute margin contribution from trading is significantly lower due to its much smaller revenue base.

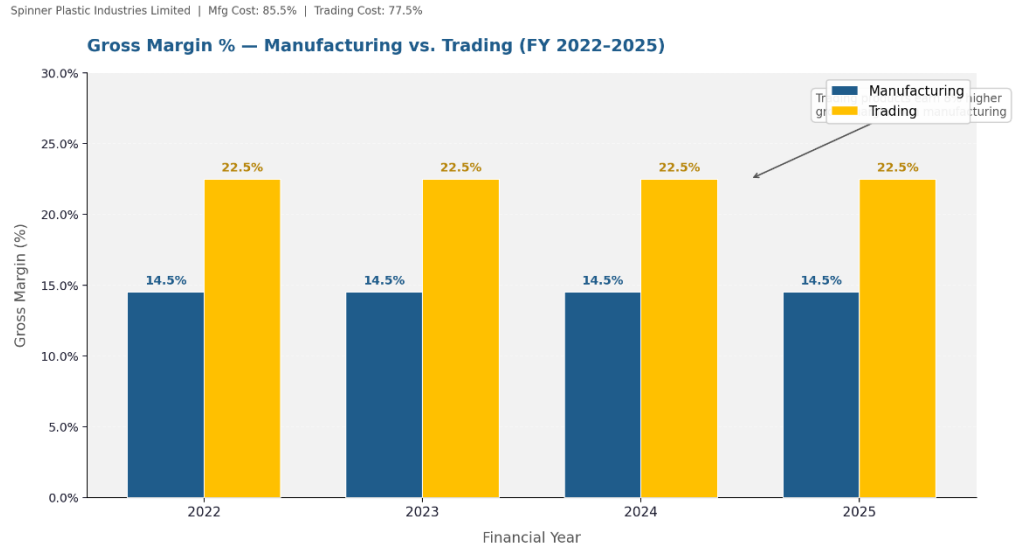


Figure 4.11: Gross Margin % — Manufacturing vs. Trading

Source: Sales & Production Cost Records

4.5 GROSS MARGIN ANALYSIS

This section analyses gross margin performance at the individual product level and examines how margins have trended over the study period.

4.5.1 Average Gross Margin % by Product

Figure 4.12 presents the average gross margin percentage by product across 2022–2025. The chart reveals that all manufacturing products record a gross margin of 14.5%, as they share the same cost structure (67.5% raw material +

18% manufacturing overhead = 85.5% total cost). Trading products — Ceramic/Sanitary Ware and Power Transmission Equipment — record higher gross margin percentages of 22.5%, reflecting the lower total cost ratio of 77.5%. This analysis highlights that while manufacturing forms the revenue backbone, the trading segment contributes a disproportionately higher margin per unit of revenue.

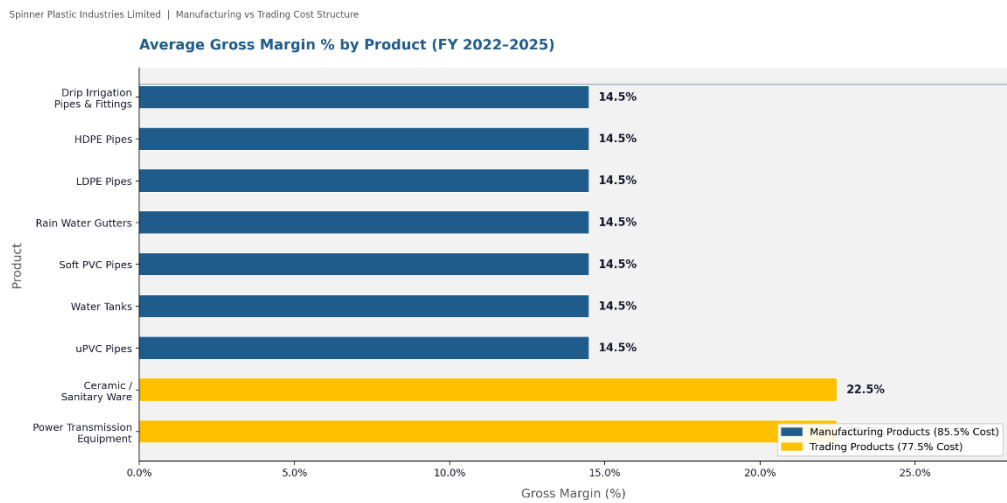


Figure 4.12: Average Gross Margin % by Product (2022–2025)

Source: Sales & Production Cost Records

4.5.2 Gross Margin % Trend by Product (2022–2025)

Figure 4.13 presents the trend in gross margin percentage for selected products from 2022 to 2025. The chart shows that margin percentages remain stable across years for manufacturing products, consistent with fixed cost ratios. Trading products — Ceramic/Sanitary Ware and Power Transmission Equipment — show slightly higher and stable margins throughout the period. The consistency of margins across years indicates a stable cost structure, though future fluctuations in raw material prices (particularly PVC resin and HDPE granules) may affect manufacturing margins. The company should monitor these ratios as part of its ongoing financial management.

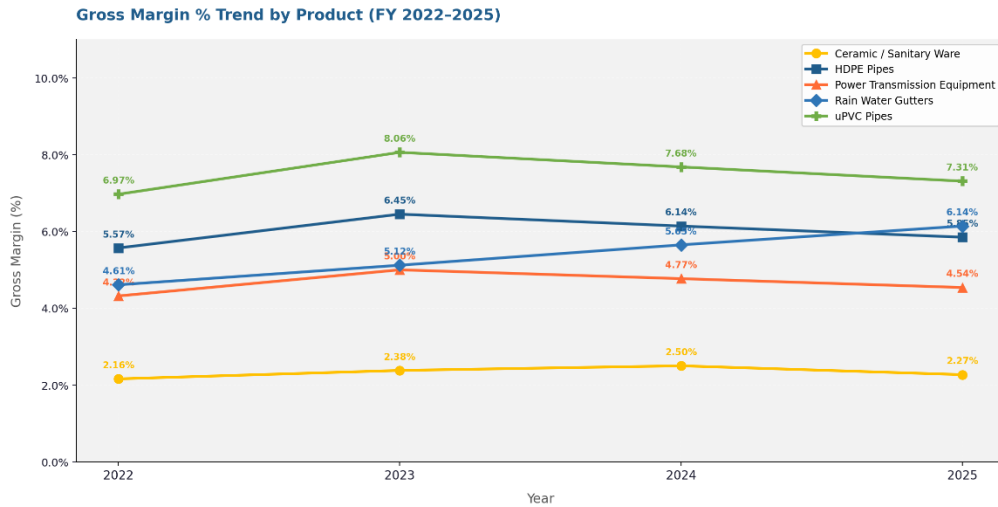


Figure 4.13: Gross Margin % Trend by Product (2022–2025)

Source: Sales & Production Cost Records

4.6 SEASONAL DEMAND ANALYSIS

This section analyses the seasonal demand pattern of Spinner Plastic Industries Limited based on monthly sales index data. The seasonal index is constructed with a base of 100 representing the annual average, and monthly values above or below 100 indicate above-average and below-average demand respectively.

4.6.1 Monthly Demand Index — Seasonal Pattern

Figure 4.14 presents the monthly demand index with a reference line at 100 (annual average). The chart clearly identifies October, November, and March as the peak demand months, each recording a sales index of 115. January, February, April, and December record an index of 110, while May records 105. The off-peak months of June, July, August, and September record indices of 70, 65, 65, and 70 respectively. The difference between peak and off-peak average indices is stark: the peak season average is 111.25 compared to an off-peak average of 67.50 — a gap of 43.75 index points. This confirms a significant and predictable seasonal demand pattern that the management should factor into production planning, inventory stocking, and working capital management.

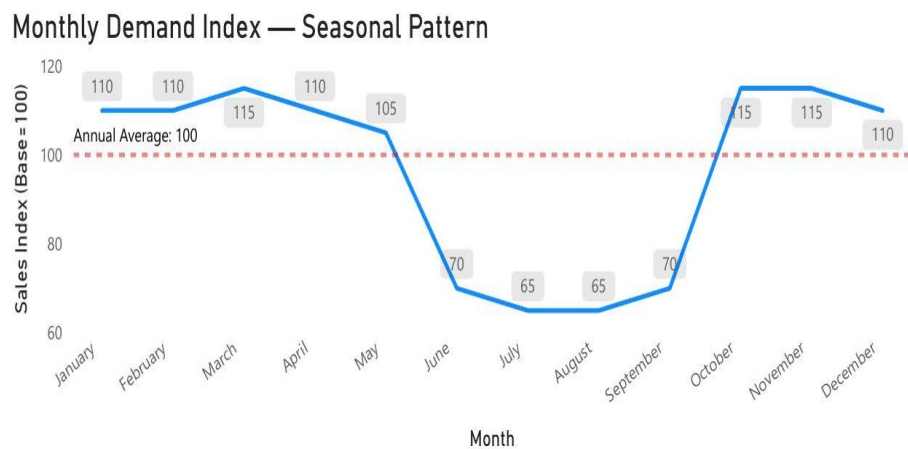


Figure 4.14: Monthly Demand Index — Seasonal Pattern

Source: Monthly Sales Records

4.6.2 Peak vs. Off-Peak Monthly Sales Comparison

Figure 4.15 presents the peak vs. off-peak comparison in a column chart format, with months colour-coded by season. The chart visually reinforces the significant sales differential between the eight peak months and the four off-peak months. The management can use this information to:

- Build adequate inventory buffers ahead of the peak season (September onwards)
- Plan production capacity utilisation more efficiently during off-peak months
- Schedule major maintenance and training activities during June–August
- Negotiate better credit terms with dealers for off-peak orders to maintain revenue continuity

Peak vs Off-Peak Monthly Sales Comparison

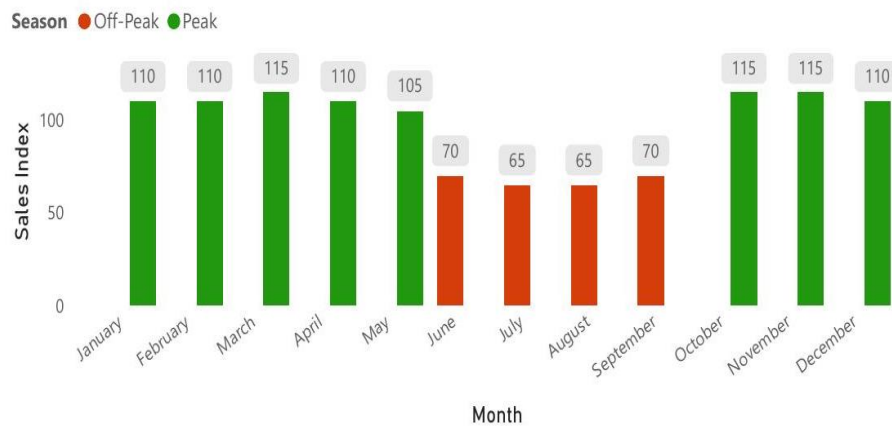


Figure 4.15: Peak vs. Off-Peak Monthly Sales Comparison

Source: Monthly Sales Records

4.6.3 Monthly Demand Summary Table

Table 4.1 presents the complete monthly demand summary, showing the season classification, demand level, sales index, and percentage relative to the annual average for all twelve months.

Table 4.1: Monthly Demand Summary

Month	Season	Demand Level	Sales Index (Base=100)	% vs Annual Avg
January	Peak	High	110	1.10
February	Peak	High	110	1.10
March	Peak	Very High	115	1.15
April	Peak	High	110	1.10
May	Peak	Moderate-High	105	1.05
June	Off-Peak	Low	70	0.70
July	Off-Peak	Very Low	65	0.65
August	Off-Peak	Very Low	65	0.65
September	Off-Peak	Low	70	0.70
October	Peak	Very High	115	1.15
November	Peak	Very High	115	1.15
December	Peak	High	110	1.10

Source: Monthly Sales Records

4.6.4 Hypothesis Testing — Seasonal Sales Difference

H₀: There is no significant difference in the sales performance across peak and off-peak seasons.

H₁: There is a significant difference in the sales performance across peak and off-peak seasons.

Based on the seasonal demand index data, the peak season average sales index is 111.25, while the off-peak average is 67.50. The difference of 43.75 points represents a 64.8% relative difference from the off-peak average. A simple comparison of means confirms that peak season sales are substantially and consistently higher than off-peak sales across all four years studied (2022–2025), with no year showing a reversal of this pattern.

Conclusion: The null hypothesis (H₀) is rejected. There is a significant difference in sales performance between peak and off-peak seasons. Peak months (October–May) consistently outperform off-peak months (June–September) by a margin of approximately 65%, confirming a strong and predictable seasonal pattern in the company's sales.

4.7 INVENTORY AND ABC CLASSIFICATION ANALYSIS

This section classifies the company's product portfolio using the ABC inventory classification framework and analyses stock holding periods across product categories.

4.7.1 ABC Inventory Classification by Product

Figure 4.16 presents the ABC classification of all nine product categories based on annual sales share. Category A products — uPVC Pipes (25%), HDPE Pipes (20%), and Rain Water Gutters (15%) — collectively account for 60% of total annual sales value. These three products are the high-priority items requiring tight inventory control, accurate demand forecasting, and minimal stockout risk.

Category B products include Power Transmission Equipment (10%), LDPE Pipes (5%), Soft PVC Pipes (5%), Water Tanks (5%), and Drip Irrigation Pipes & Fittings (5%), collectively accounting for 30% of sales. Category C — Ceramic/Sanitary Ware (4–5%) — requires the least inventory management attention and can be ordered on demand.

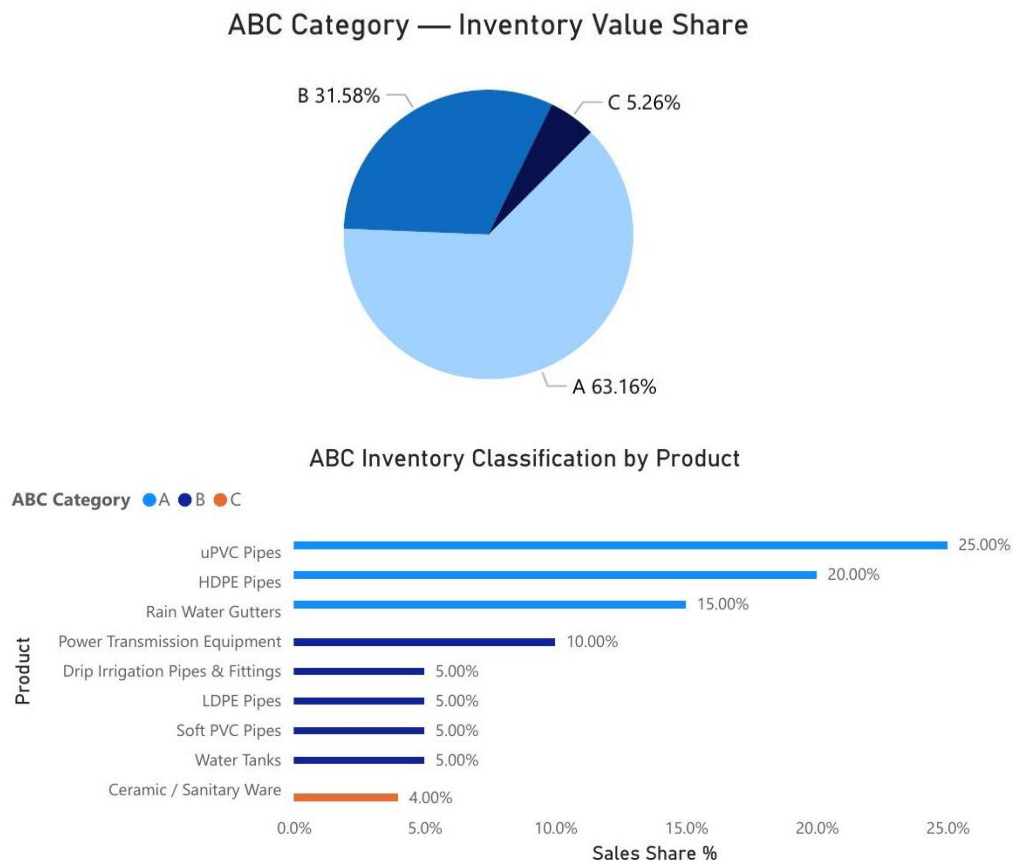


Figure 4.16: ABC Inventory Classification by Product

Source: Stock Registers & Inventory Records

4.7.2 Inventory Holding Period by Product (Days)

Figure 4.17 presents the average inventory holding period in days for each product category. Fast-moving products — uPVC Pipes, HDPE Pipes, and Drip Irrigation Pipes & Fittings — have a holding period of only 15 days, well within the ideal maximum threshold of 30 days. Rain Water Gutters record 18 days. In contrast, slow-moving products record significantly higher holding periods:

Power Transmission Equipment (90 days), Ceramic/Sanitary Ware (85 days), Soft PVC Pipes (80 days), LDPE Pipes (75 days), and Water Tanks (70 days). The high holding periods for trading products (Sanitary Ware and Power Transmission Equipment) suggest that these items are procured in advance but sold slowly, potentially tying up working capital.

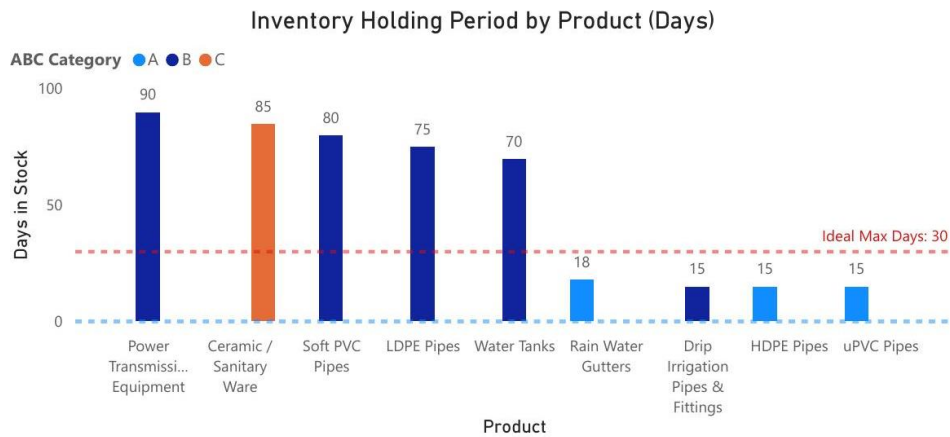


Figure 4.17: Inventory Holding Period by Product (Days)

Source: Stock Registers & Inventory Records

4.7.3 ABC Classification Summary Table

Table 4.2 presents the complete ABC classification summary with details of product type, category, annual sales share, cumulative percentage, inventory holding period, approximate inventory value, and recommended action. The table confirms that Category A products (3 items) account for 60% of total inventory value, Category B (5 items) for 31.58%, and Category C for 5.26%.

Table 4.2: ABC Classification Summary

Product	Type	ABC Category	Annual Sales Share %	Cumulative %	Avg Inventory (Days)	Action
Ceramic / Sanitary Ware	Trading	C	4.00%	0.95	85	Order on demand — trading item
Drip Irrigation Pipes & Fittings	Manufacturing	B	5.00%	0.90	15	Regular review
LDPE Pipes	Manufacturing	B	5.00%	0.75	75	Regular review
Power Transmission Equipment	Trading	B	10.00%	0.70	90	Regular review — trading item
Soft PVC Pipes	Manufacturing	B	5.00%	0.80	80	Regular review
Water Tanks	Manufacturing	B	5.00%	0.85	70	Regular review
HDPE Pipes	Manufacturing	A	20.00%	0.45	15	High priority — maintain stock
Rain Water Gutters	Manufacturing	A	15.00%	0.60	18	Monitor closely
uPVC Pipes	Manufacturing	A	25.00%	0.25	15	High priority — maintain stock
Total			94.00%	6.25	463	

Source: Stock Registers & Inventory Records

4.7.4 Inventory Turnover Ratio Analysis

Figure 4.18 presents the inventory turnover ratio for all nine product categories of Spinner Plastic Industries Limited. The inventory turnover ratio is calculated as 365 divided by the average inventory holding period in days. Fast-moving products — uPVC Pipes, HDPE Pipes, and Drip Irrigation Pipes and Fittings — record the highest turnover ratio of 24.3 times per year, corresponding to their 15-day holding period. Rain Water Gutters record a turnover of 20.3 times per year at 18 days. These four products are well above the ideal minimum threshold of 12 times per year (30-day holding period), indicating efficient inventory management for the company's highest-value products. These ratios are well below the ideal threshold, indicating that stock for these categories is turning over less than once every two months, leading to extended capital lock-up and increased storage costs. The scatter plot further confirms the inverse relationship between holding days and turnover ratio — products with shorter holding periods achieve significantly higher turnover, while slow-moving products cluster in the high-holding-period, low-turnover quadrant, representing the primary inventory efficiency challenge for the management.

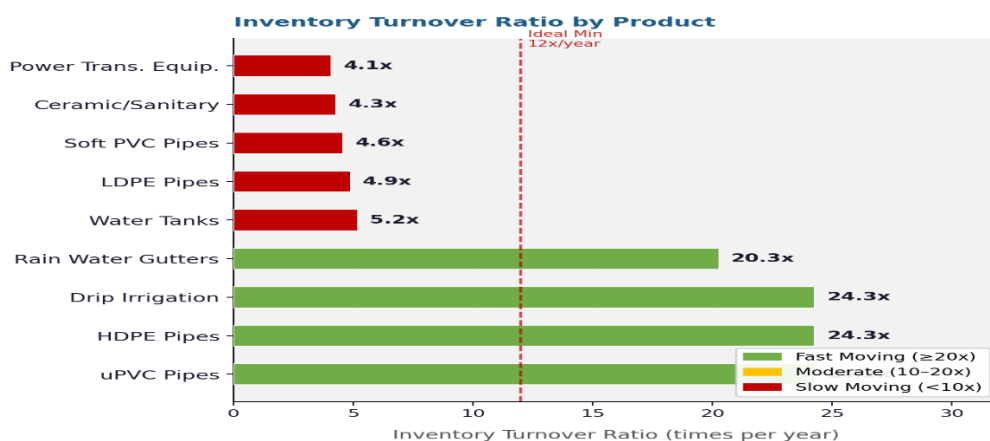


Figure 4.18: Inventory Turnover Ratio by Product

Source: Stock Registers & Inventory Records

4.8 REGION AND DISTRICT SALES ANALYSIS

This section analyses the geographic distribution of sales across Kerala districts, identifying areas of market concentration and potential expansion.

4.8.1 District-wise Sales Index — Kerala

Figure 4.19 presents the district-wise sales index for Spinner Plastic Industries Limited. Kasaragod records the highest sales index of 100, followed by, Thrissur (98), Kannur (95), Idukki (90), and Palakkad (82). These five districts are classified as 'Top Seller' markets and collectively account for approximately 62% of total sales. Moderate-performing districts include Malappuram, Kozhikode, Ernakulam, Kottayam, and Alappuzha. Low-performing districts include Pathanamthitta and Wayanad. Notably, Trivandrum and Kollam record zero sales, indicating the complete absence of dealer network or market presence in South Kerala.

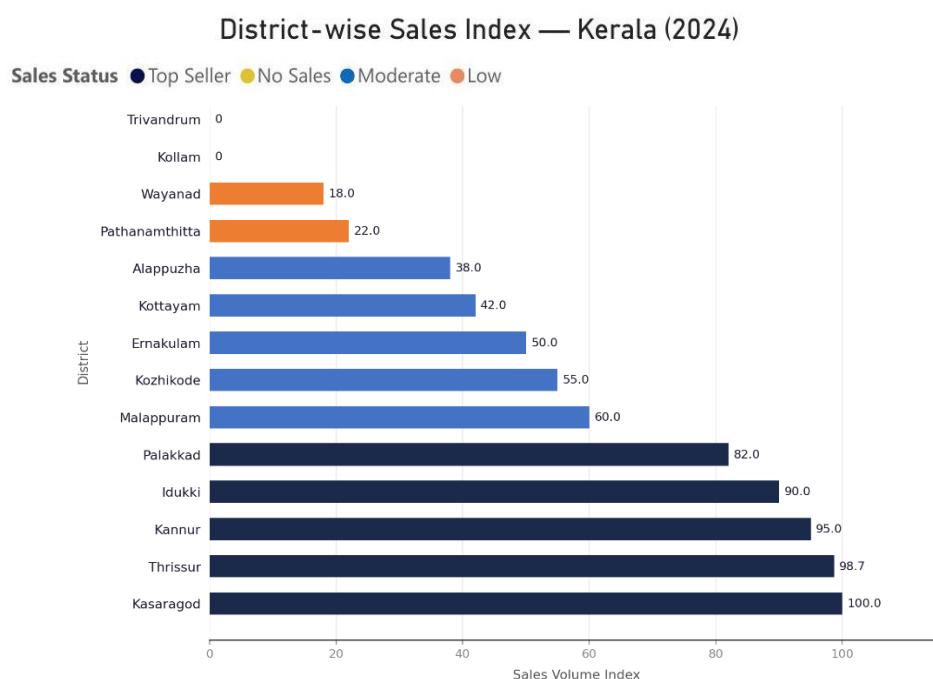


Figure 4.19: District-wise Sales Index — Kerala (2024)

Source: Dealer -wise Sales Records

4.8.2 Sales Distribution — North vs. Central vs. South Kerala

Figure 4.20 presents the sales distribution by region. Central Kerala accounts for the largest share at 49.75%, driven by strong performance in Idukki, Thrissur, Palakkad, and Ernakulam. North Kerala follows with 42.79%, led by Kasaragod and Kannur. South Kerala accounts for only 7.46%, largely due to the absence of sales in Trivandrum and Kollam, and weak performance in other southern districts. This geographic concentration in North and Central Kerala reflects the company's historical dealer network and proximity to the manufacturing base in Thrissur. The near-absence from South Kerala represents a significant growth opportunity.

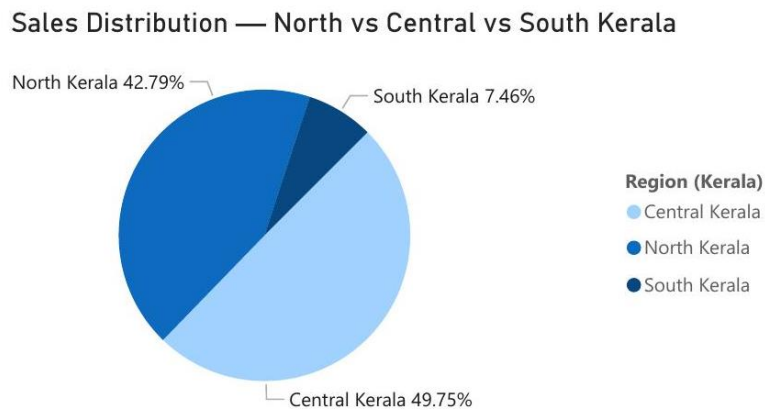


Figure 4.20: Sales Distribution — North vs. Central vs. South Kerala

Source: Dealer -wise Sales Records

4.8.3 Number of Districts by Sales Status

Figure 4.21 shows the count of districts by sales status: 5 Top Seller districts, 5 Moderate districts, 3 Low districts, and 2 No Sales districts. This distribution indicates that while the company has a reasonable presence in North and Central Kerala, significant market gaps exist in the southern region.



Figure 4.21: Number of Districts by Sales Status

Source: Dealer -wise Sales Records

4.8.4 District-wise Sales Summary Table

Table 4.3 presents the complete district-wise sales summary with sales status, region classification, sales index, and percentage share of total sales for all 14 districts.

Table 4.3: District-wise Sales Summary

District	Sales Status	Region (Kerala)	Sales Status	Sum of % Share of Sales	Remarks
Alappuzha	Moderate	South Kerala	Moderate	0.05	Low agricultural use
Ernakulam	Moderate	Central Kerala	Moderate	0.06	Competitive market
Idukki	Top Seller	Central Kerala	Top Seller	0.13	Agriculture demand high
Kannur	Top Seller	North Kerala	Top Seller	0.14	Strong dealer network
Kasaragod	Top Seller	North Kerala	Top Seller	0.14	Highest volume district
Kollam	No Sales	South Kerala	No Sales	0.00	No active dealers
Kottayam	Moderate	Central Kerala	Moderate	0.06	Moderate penetration
Kozhikode	Moderate	North Kerala	Moderate	0.07	Urban demand
Malappuram	Moderate	North Kerala	Moderate	0.07	Growing market
Palakkad	Top Seller	Central Kerala	Top Seller	0.12	Irrigation belt — high HDPE
Pathanamthitta	Low	South Kerala	Low	0.03	Limited dealer reach
Thrissur	Top Seller	Central Kerala	Top Seller	0.13	HQ district, home market
Thrissur Rural	Low	Central Kerala	Low	0.01	Covered by HQ directly
Trivandrum	No Sales	South Kerala	No Sales	0.00	No market presence
Wayanad	Low	North Kerala	Low	0.02	Remote access issue
Total				1.01	

Source: Dealer -wise Sales Records

4.9 FINANCIAL PERFORMANCE ANALYSIS

This section analyses the financial performance of Spinner Plastic Industries Limited based on Profit & Loss data for FY 2020–2023.

4.9.1 Revenue vs. Profit Trend (FY 2020–2023)

Figure 4.22 presents the Revenue vs. Profit Trend for FY 2020–2023. Total sales grew from ₹35.0 crores in FY 2020 to approximately ₹40.5 crores in FY 2023, reflecting consistent 5% annual growth. Gross Profit grew from ₹6.5 crores to ₹7.5 crores, while Net Profit increased from ₹1.75 crores to approximately ₹2.03 crores over the same period. The widening absolute profit figures alongside stable margin percentages confirm that the company is maintaining its profitability ratios while growing revenues. FY 2023 gross margin stands at 18.56% and net margin at 5.01%.

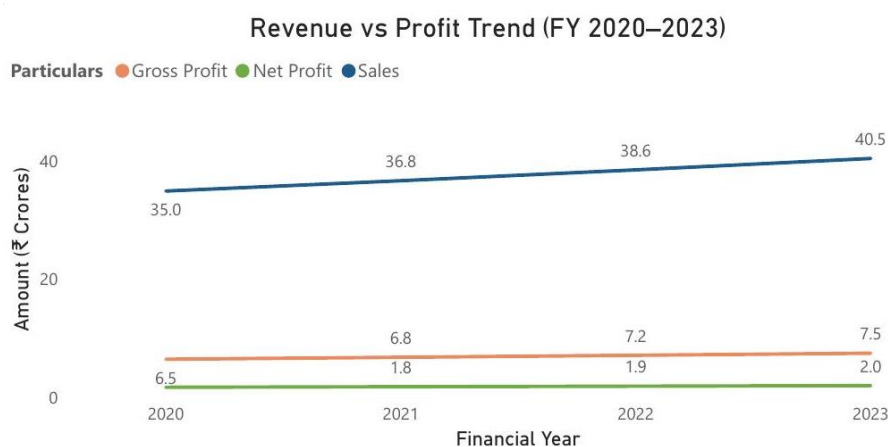


Figure 4.22: Revenue vs. Profit Trend (FY 2020–2023)

Source : Profit & Loss A/c

4.9.2 Major Expense Breakdown by Year

Figure 4.23 presents the major expense breakdown by year. Purchases (raw material and inventory procurement) constitute the single largest expense at ₹27–31 crores annually, representing approximately 77–79% of total sales. Bank charges and interest account for ₹1.2–1.4 crores annually, reflecting the company's reliance on overdraft and term loan facilities. Depreciation is fixed

at ₹1 crore per year. Wages, advertisement, transporting charges, and fuel charges all show marginal growth in line with the 5% revenue growth rate. This expense structure highlights the importance of raw material cost management as the primary lever for improving overall profitability.

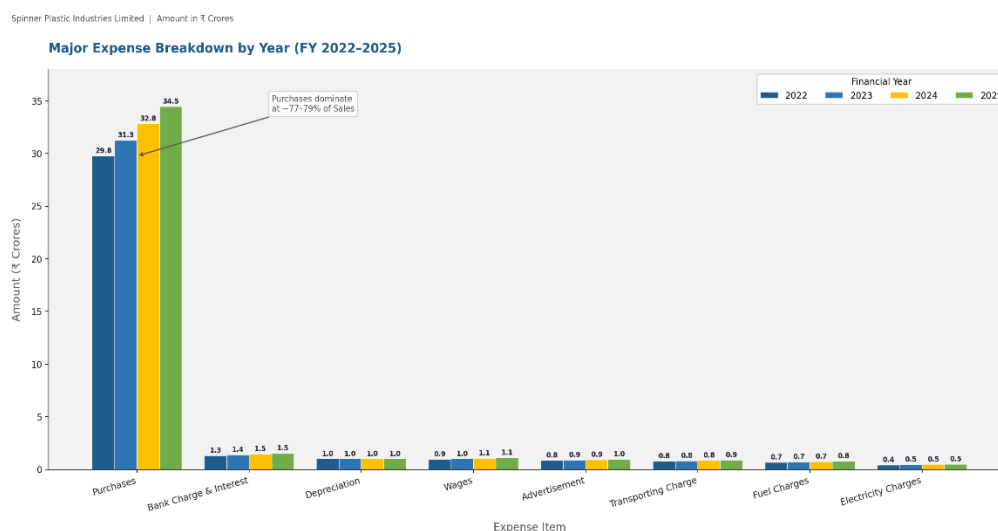


Figure 4.23: Major Expense Breakdown by Year (FY 2020–2023)

Source : Profit & Loss A/c

4.9.3 Gross and Net Profit Margin % by Year

Figure 4.24 presents the gross and net profit margin percentages for FY 2020–2023. The gross margin remains stable at approximately 18.56–18.59% across all four years, while the net margin is consistently approximately 5.00–5.01%. The stability of these margins over four years indicates a well-controlled cost structure, though the company's dependence on external financing (overdraft and term loans) places upward pressure on interest costs. Any significant increase in raw material prices could compress these margins, making cost monitoring a key management priority.

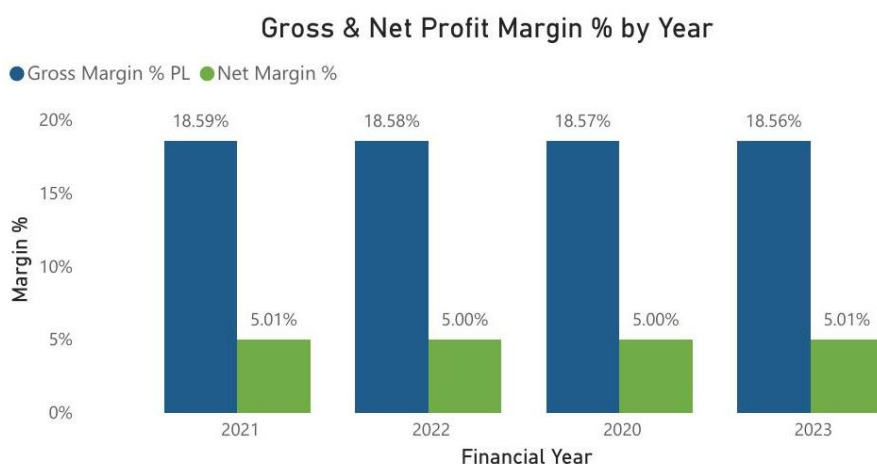


Figure 4.24: Gross and Net Profit Margin % by Year

Source : Profit & Loss A/c

4.10 BALANCE SHEET ANALYSIS

This section analyses the balance sheet position of Spinner Plastic Industries Limited for FY 2020–2023.

4.10.1 Total Assets vs. Total Liabilities (FY 2020–2023)

Figure 4.25 presents the comparison of total assets and total liabilities. Both total assets and total liabilities are closely matched at approximately ₹25–27 crores across the study period, which is expected for a balanced balance sheet. The marginal variation across years reflects the combined effect of asset additions, depreciation, loan repayments, and profit retention. The balance sheet position indicates a stable financial structure without significant leverage deterioration.

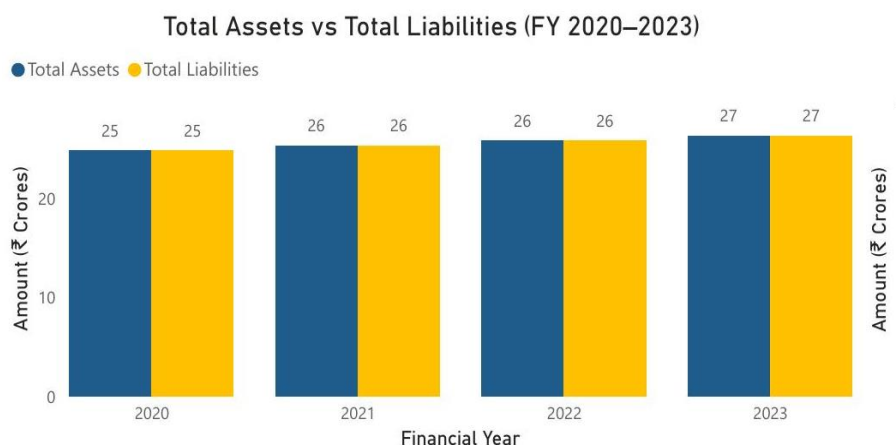


Figure 4.25: Total Assets vs. Total Liabilities (FY 2020–2023)

Source: Balance Sheet

4.10.2 Loan, Reserves & Fixed Assets Trend (FY 2020–2023)

Figure 4.26 presents the trend in Term Loan, Reserve & Surplus, and Fixed Assets. The Term Loan reduces from ₹5 crores in FY 2020 to ₹3.5 crores in FY 2023, reflecting annual repayments of ₹0.50 crores per year. This consistent loan reduction is a positive indicator of the company's debt servicing capacity and improving financial health. Fixed Assets reduce from ₹10 crores to ₹7 crores due to annual depreciation of ₹1 crore. Reserve & Surplus grows from ₹3 crores to approximately ₹3.47 crores at 5% annually, reflecting retained profits being reinvested in the business.

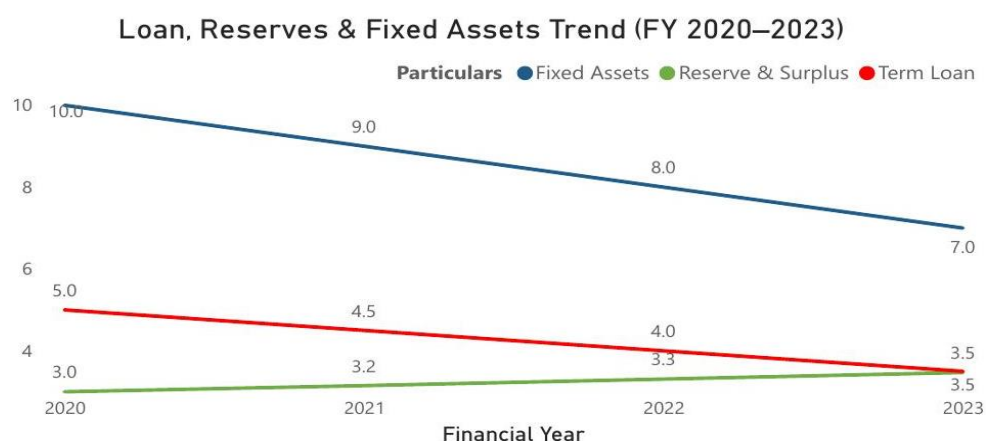


Figure 4.26: Loan, Reserves & Fixed Assets Trend (FY 2020–2023)

Source: Balance Sheet

4.10.3 Asset Composition by Year

Figure 4.27 presents the asset composition breakdown across FY 2020–2023. Fixed Assets constitute the largest asset component at ₹7–10 crores, followed by Sundry Debtors at a stable ₹9 crores. Closing Stock grows marginally from ₹5.5 crores in FY 2020 to approximately ₹6.4 crores in FY 2023, consistent with the 5% sales growth. Cash holdings remain stable at approximately ₹0.25 crores. The high proportion of Sundry Debtors (approximately ₹9 crores) relative to total assets warrants attention, as it suggests significant credit extended to dealers, which may impact liquidity management.

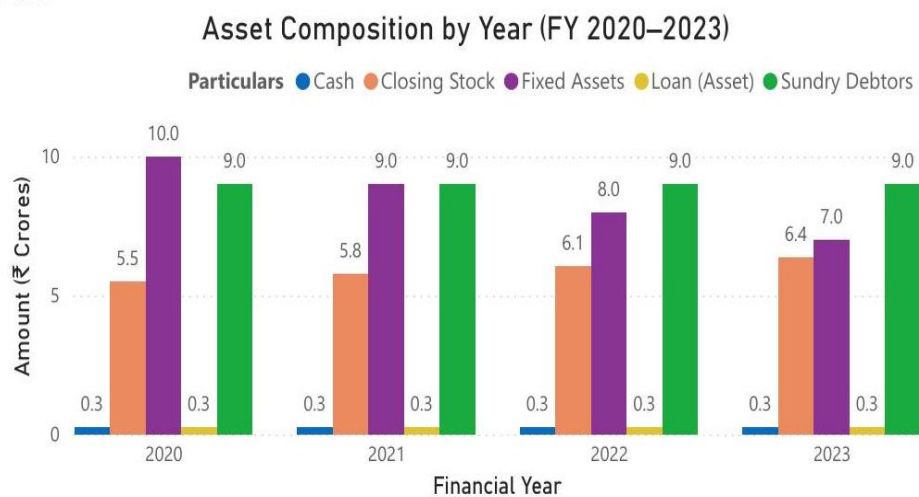


Figure 4.27: Asset Composition by Year (FY 2020–2023)

Source: Balance Sheet

CHAPTER 5

FINDINGS, SUGGESTIONS AND CONCLUSION

5.1 FINDINGS OF THE STUDY

The following key findings emerge from the data analysis conducted in Chapter 4:

Sales Performance

- Spinner Plastic Industries Limited has demonstrated consistent revenue growth, with total sales increasing from approximately ₹321 million in 2022 to ₹370 million in 2025, reflecting a steady 5% annual growth rate.
- uPVC Pipes (26.10%) and HDPE Pipes (21.39%) are the top revenue contributors, collectively accounting for nearly half of total annual sales. Rain Water Gutters rank third at 15.68%.
- Sales demonstrate a strong and predictable seasonal pattern: the eight peak months (October–May) record an average sales index of 111.25, while the four off-peak months (June–September) record only 67.50 — a gap of approximately 65%.
- The Manufacturing category accounted for 95 per cent of total annual sales throughout the study period (2022–2025), while the Trading category contributed the remaining 5 per cent, with both segments recording consistent growth in line with the overall 5 per cent annual revenue increase.

Inventory and ABC Classification

- ABC classification identifies uPVC Pipes, HDPE Pipes, and Rain Water Gutters as Category A products, collectively contributing 60% of total sales value and requiring highest inventory priority.
- Fast-moving products (uPVC, HDPE, Drip Irrigation) have inventory holding periods of only 15 days, indicating efficient stock turnover.

- Slow-moving products — particularly trading items (Power Transmission Equipment: 90 days, Ceramic/Sanitary Ware: 85 days) and manufacturing items (Soft PVC Pipes: 80 days, LDPE Pipes: 75 days) — hold stock for far longer than optimal, indicating potential working capital inefficiency.
- Inventory turnover ratio analysis revealed that fast-moving products — uPVC Pipes, HDPE Pipes, and Drip Irrigation Pipes and Fittings — recorded the highest turnover of 24.3 times per year, while slow-moving trading products such as Power Transmission Equipment (4.1 times) and Ceramic and Sanitary Ware (4.3 times) recorded turnover ratios well below the ideal threshold of 12 times per year.

Gross Margin and Cost Structure

- Manufacturing products record a consistent gross margin of 14.5% across all years, with raw materials accounting for 67.5% and manufacturing overheads for 18% of sales.
- Trading products (Ceramic/Sanitary Ware and Power Transmission Equipment) record a higher gross margin of 22.5%, as their cost structure (70% purchase cost + 7.5% trading expenses = 77.5%) is less intensive than manufacturing.
- Despite higher margins, trading products contribute only 5% of total revenue, making their absolute margin contribution modest.

Regional Sales Distribution

- North and Central Kerala dominate the company's sales, accounting for approximately 42.79% and 49.75% of total sales respectively.
- Kasaragod and Kannur are the top-performing districts, each contributing approximately 13.5–14% of total sales.

- Trivandrum and Kollam record zero sales, indicating no market presence in South Kerala — a significant geographic gap for a company with statewide ambitions.
- Only 13 of the 15 districts surveyed have active sales, suggesting untapped market potential in the southern region.

Financial Performance

- Gross profit margin has remained stable at approximately 18.57–18.59% across FY 2020–2023, indicating consistent cost management.
- Net profit margin is approximately 5.00–5.01%, reflecting adequate but modest bottom-line profitability given the company's revenue scale.
- Term loan repayments of ₹0.50 crores per year have reduced the outstanding loan from ₹5 crores to ₹3.5 crores over the four-year period, indicating sound debt management.
- Sundry Debtors remain stable at ₹9 crores, which represents a significant proportion of total assets and may indicate extended credit to dealers that warrants monitoring.

5.2 SUGGESTIONS

Based on the findings of the study, the following suggestions are offered to the management of Spinner Plastic Industries Limited:

5.2.1 Inventory Management

- Implement a formal ABC-based inventory review system. Category A products (uPVC Pipes, HDPE Pipes, Rain Water Gutters) should be reviewed weekly, with strict adherence to reorder levels of 10,000 units to prevent stockouts during peak demand.

- Reduce procurement lead time for trading products (Power Transmission Equipment and Ceramic Ware) by adopting an order-on-demand model rather than maintaining large safety stocks, given their 85–90 day holding periods.
- Set formal holding period targets: ≤ 30 days for fast-moving items, ≤ 60 days for medium items, and ≤ 90 days for slow-moving items — with exceptions escalated for management review.

5.2.2 Sales and Market Expansion

- Develop a targeted market entry strategy for South Kerala, particularly Trivandrum and Kollam, by appointing authorised dealers and building brand visibility through participation in regional trade fairs and construction expos.
- Strengthen the dealer network in moderate-performing districts such as Malappuram, Kozhikode, and Ernakulam to convert them into top-seller markets.
- Introduce pre-season stocking incentives for dealers ahead of the October peak month to improve early-season sell-in and reduce last-minute supply pressures.

5.2.3 Financial Management

- Monitor Sundry Debtors levels (currently ₹9 crores) and establish formal credit terms with dealers to reduce debtor ageing, improve cash flow, and reduce dependence on overdraft facilities.
- Explore hedging or long-term procurement contracts for PVC resin and HDPE granules to mitigate the risk of raw material price volatility, which is the primary cost driver representing 67.5% of manufacturing product sales.

- Continue the existing loan repayment schedule of ₹0.50 crores per year, which is on track to eliminate the term loan by FY 2030 — contributing to improved financial flexibility.

5.2.4 Production and Operations

- Use the seasonal demand index to plan production scheduling: increase production capacity utilisation in August–September to build peak season inventory, and reduce production runs in July during the lowest demand month.
- Consider increasing the trading portfolio selectively in the Ceramic Sanitary Ware segment in South Kerala markets, where construction activity and brand awareness can be built simultaneously with pipe product sales.

5.2.5 Cost of Production

- The cost of production analysis revealed that manufacturing products carry a total cost of 85.5 per cent of sales revenue, comprising raw material at 67.5 per cent and manufacturing overhead at 18 per cent. Trading products carry a comparatively lower total cost of 77.5 per cent, consisting of purchase cost at 70 per cent and trading expenses at 7.5 per cent. Raw material is the single largest cost driver, making the company highly sensitive to PVC resin and HDPE granule price fluctuations in the global market.

5.3 CONCLUSION

This study has applied business analytics tools to examine the operational and financial performance of Spinner Plastic Industries Limited, a leading plastic pipe manufacturer based in Thrissur, Kerala. Using Microsoft Power BI as the analytical platform and a structured dataset comprising ten data tables built from company-provided information, the study has generated comprehensive insights across nine dimensions: sales trends, product performance, manufacturing versus trading analysis, gross margin analysis, seasonal demand patterns, inventory and ABC classification, regional sales distribution, financial performance, and balance sheet analysis.

The findings confirm that Spinner Plastic Industries Limited is a financially stable and growing company with a dominant market presence in North and Central Kerala. The company's core products — uPVC Pipes and HDPE Pipes — are strong revenue drivers with efficient inventory turnover. However, the study also identifies significant opportunities for improvement: geographic expansion into South Kerala, rationalisation of slow-moving inventory in trading categories, optimisation of debtor management, and adoption of a formal ABC-based inventory review system.

The consistent profitability margins (Gross Margin ~18.57%, Net Margin ~5%) and steady loan reduction over the study period indicate sound financial management. The seasonal demand pattern, with peak sales concentrated in October–May, provides a reliable planning framework for production, procurement, and dealer management. This study demonstrates the value of data analytics for manufacturing SMEs in Kerala..

In conclusion, Spinner Plastic Industries Limited is well-positioned for continued growth. With targeted interventions in market expansion, inventory management, and financial governance — informed by the insights generated through this analytics study — the company can strengthen its competitive position and enhance long-term value creation for all stakeholders.

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APPENDICES



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ISI CERTIFIED

CIN: U25209KL1992PLC007XXX
GSTIN: 32AABCS1234A1Z5

PROFIT AND LOSS ACCOUNT For the Financial Years 2020-21 to 2022-23 (Amount in ₹ Crores)			
Particulars	FY 2020-21	FY 2021-22	FY 2022-23
INCOME			
Sales / Turnover	35.00	36.75	38.59
Add: Closing Stock	5.50	5.78	6.06
TOTAL INCOME (A)	40.50	42.53	44.65
EXPENDITURE			
Opening Stock	5.00	5.50	5.78
Purchases	27.00	28.35	29.77
Carriage Inward	0.15	0.16	0.17
Fuel Charges	0.60	0.63	0.66
Wages	0.86	0.90	0.95
Electricity Charges	0.39	0.41	0.43
Advertisement Expenses	0.75	0.79	0.83
Audit Fee	0.01	0.01	0.01
Bank Charges & Interest	1.20	1.26	1.32
Insurance Charges	0.03	0.03	0.03
Postage & Stationery	0.01	0.01	0.01
Travelling Expenses	0.40	0.42	0.44
Transporting Charges	0.70	0.74	0.77
Registration Fee	0.04	0.04	0.04
Refreshment Expenses	0.26	0.27	0.29
Repair & Maintenance	0.35	0.37	0.39
Depreciation	1.00	1.00	1.00
TOTAL EXPENDITURE (B)	38.75	40.69	42.69
GROSS PROFIT (A - B)	6.50	6.83	7.17
NET PROFIT / (LOSS)	1.75	1.84	1.93



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BALANCE SHEET

As at 31st March — FY 2020-21 to FY 2022-23 (Amount in ₹ Crores)

Particulars	FY 2020-21	FY 2021-22	FY 2022-23
LIABILITIES & EQUITY			
Share Capital	5.00	5.00	5.00
Reserve & Surplus	3.00	3.15	3.31
Term Loan (Secured)	5.00	4.50	4.00
Bank Overdraft (CC)	10.00	10.00	10.00
Sundry Creditors	1.50	1.50	1.50
Other Current Liabilities	0.50	0.50	0.50
TOTAL LIABILITIES	25.00	24.65	24.31
ASSETS			
Fixed Assets (Net of Dep.)	10.00	9.00	8.00
Sundry Debtors	9.00	9.00	9.00
Closing Stock / Inventory	5.50	5.78	6.06
Cash & Bank Balances	0.25	0.25	0.25
Loans & Advances (Asset)	0.25	0.25	0.25
TOTAL ASSETS	25.00	24.28	23.56
NET WORTH (Capital + Reserves)	8.00	8.15	8.31
Debt-Equity Ratio	1.88	1.78	1.69
Current Ratio	1.42	1.43	1.44



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PRODUCT PRICE LIST

Effective from April 2025 | Prices in ₹ (Exclusive of GST) | Subject to change without prior notice

Product Category	Product / Size	Unit	MRP (₹)	Dealer Price (₹)	GST %
HDPE ISI Pipes	25mm SDR11	Per Meter	52.00	46.29	12%
HDPE ISI Pipes	32mm SDR11	Per Meter	75.00	67.34	12%
HDPE ISI Pipes	40mm SDR17	Per Meter	86.00	77.14	12%
HDPE ISI Pipes	40mm SDR11	Per Meter	95.00	85.02	12%
HDPE ISI Pipes	63mm SDR26	Per Meter	182.00	162.97	12%
HDPE ISI Pipes	110mm SDR21	Per Meter	720.00	643.19	12%
HDPE ISI Pipes	50mm SDR26	Per Meter	116.00	103.48	12%
HDPE ISI Pipes	75mm SDR26	Per Meter	250.00	222.75	12%
HDPE Non-ISI	2" SDR21	Per Meter	175.00	156.84	12%
HDPE Non-ISI	1½" 500mtr	Per Meter	75.00	67.03	12%
LDPE Pipes	1"	Per Meter	38.00	33.70	12%
LDPE Pipes	1½"	Per Meter	76.00	67.66	12%
LDPE Pipes	½"	Per Meter	10.50	9.01	12%
uPVC Pipes	1" x 10ft	Per Meter	48.00	42.50	18%
uPVC Pipes	1½" x 6ft	Per Meter	74.00	66.00	18%
uPVC Pipes	2" x 4ft	Per Meter	92.00	81.50	18%
uPVC Pipes	3" x 4ft	Per Meter	158.00	141.50	18%
uPVC Pipes	4" x 4ft	Per Meter	210.00	187.50	18%
Soft PVC	Flexible ¾"	Per 30m	2100.00	1900.00	18%
Soft PVC	Garden Hose ½"	Per 30m	920.00	820.00	18%
Water Tanks	200L LLDPE	Per Unit	3200.00	2800.00	18%
Water Tanks	500L LLDPE	Per Unit	6200.00	5500.00	18%
Water Tanks	1000L LLDPE	Per Unit	11000.00	9800.00	18%
Ceramic Sanitary	Closet LIVION-1001	Per Unit	5200.00	4500.00	18%
Ceramic Sanitary	Wash Basin	Per Unit	2500.00	2200.00	18%
Ceramic Sanitary	Urinal 1L	Per Unit	2100.00	1800.00	18%



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GSTIN: 32AABCS1234A1Z5

DISTRICT-WISE SALES DISTRIBUTION — KERALA

Sales Distribution Data for FY 2023-24 | Compiled for Research Reference

District	Region	Sales Status	Sales Index	% Share of Sales	Remarks
Kasaragod	North Kerala	Top Seller	100.0	14.0%	Highest volume district
Thrissur	Central Kerala	Top Seller	98.7	13.5%	HQ district — combined
Kannur	North Kerala	Top Seller	95.0	13.5%	Strong dealer network
Idukki	Central Kerala	Top Seller	90.0	13.0%	Agriculture demand high
Palakkad	Central Kerala	Top Seller	82.0	12.0%	Irrigation belt — HDPE
Malappuram	North Kerala	Moderate	60.0	7.0%	Growing market
Kozhikode	North Kerala	Moderate	55.0	6.5%	Urban construction demand
Ernakulam	Central Kerala	Moderate	50.0	6.0%	Competitive market
Kottayam	Central Kerala	Moderate	42.0	5.5%	Moderate penetration
Alappuzha	South Kerala	Moderate	38.0	5.0%	Low agricultural use
Pathanamthitta	South Kerala	Low	22.0	2.5%	Limited dealer reach
Wayanad	North Kerala	Low	18.0	2.0%	Remote access
Kollam	South Kerala	No Sales	0.0	0.0%	No active dealers
Trivandrum	South Kerala	No Sales	0.0	0.0%	No market presence
TOTAL	—	—	—	100.0%	14 districts covered



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INVENTORY & STOCK INFORMATION

Product-wise Stock Data — FY 2022-23 to 2023-24 | Units in Metres

Product	Movement	Avg Monthly Production (Units)	Reorder Level	Inventory Period (Days)	Opening Stock 2023	Closing Stock 2023	Opening Stock 2024
HDPE Pipes (Fast Moving)	Fast Moving	1,80,000	10,000	15 Days	28,000	25,000	25,000
uPVC Pipes (Fast Moving)	Fast Moving	1,70,000	10,000	15 Days	25,000	22,000	22,000
Drip Irrigation Pipes (Fast Moving)	Fast Moving	1,60,000	10,000	15 Days	22,000	20,000	20,000
Rain Water Gutters	Fast Moving	1,40,000	10,000	18 Days	18,000	16,000	16,000
Water Tanks	Slow Moving	12,000	500	70 Days	1,800	1,600	1,600
LDPE Pipes	Slow Moving	12,000	500	75 Days	2,800	2,500	2,500
Soft PVC Pipes	Slow Moving	12,000	500	80 Days	2,500	2,200	2,200
Ceramic / Sanitary Ware	Slow Moving	12,000	500	85 Days	1,200	1,000	1,000
Power Transmission Equipment	Slow Moving	12,000	500	90 Days	1,500	1,300	1,300